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Feasibility and Simulation of CO₂ Storage in Offshore Lebanon: Saline Aquifers Case Study

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Abstract

This paper presents a case study of offshore Lebanon to evaluate the influence of key geological parameters, namely porosity, permeability, and salinity, on the performance of CO₂ sequestration in marine saline aquifers. Through numerical simulations, this study investigates the long-term storage capacity of local formations, supported by a comprehensive feasibility and economic analysis. The objective is to assess the viability of implementing a CCUS project in the region, with the ultimate goal of formulating a technically sound and cost-effective storage strategy tailored to Lebanon's offshore context.

This study simulated CO₂ injection into saline aquifers across various geological settings. A representative cross-section from offshore Lebanon was utilized as the foundation for the modeling scenarios, incorporating salinity values between 35,000 and 100,000 ppm, permeability ranging from 100 to 1000 mD, and porosity between 20% and 30%. Each scenario was run over a 5,000-year timespan to evaluate CO₂ behavior with respect to sequestration mechanisms and dissolution rates. The model accounts for realistic subsurface conditions, integrating rock-fluid interactions, hydrostatic pressure equilibrium, and well design parameters for both injection and monitoring. Geological analysis, coupled with detailed calculations and sensitivity studies, was conducted to guide the optimal site selection and inform key technical decisions for offshore implementation. In addition, a feasibility study was conducted to evaluate the integration of major CO₂ emission sources along Lebanon's coastal zone with the proposed offshore storage site and to assess the technical and economic viability of transport infrastructure and pipeline routing.

The simulation results revealed a strong correlation between the formation properties and overall storage efficiency. High permeability was found to compromise retention and increase the risk of upward migration, whereas higher porosity enhanced CO₂ solubility and storage potential. Given the inverse relationship between salinity and CO₂ solubility, formations with lower salinity were more favorable for long-term storage. Among the evaluated cases, the scenario characterized by 20% porosity, 100 mD permeability, and 35,000 ppm salinity offered the most balanced performance, combining effective retention with efficient convective dissolution. In contrast, the scenario with 30% porosity, 1,000 mD permeability, and 100,000 ppm salinity yielded the highest initial storage capacity, but raised concerns regarding solubility limitations and potential leakage over extended timescales.

Block 4 in Lebanon's offshore domain was identified as a promising site, owing to its favorable geological characteristics—overlain by a thick evaporitic seal—and its proximity to major CO₂-emitting power plants. The total project cost was estimated at approximately ±\$500 million, incorporating expenditures related to pipeline infrastructure, labor, and ancillary services. Environmental assessments have indicated the potential for significant reductions in national greenhouse gas emissions, improvements in ambient air quality, and the creation of employment opportunities. The feasibility analysis confirmed that under appropriate assumptions for CO₂ credit pricing, CAPEX, and OPEX, the project can achieve economic viability. Thus, the proposed initiative offers a scalable and strategic framework for advancing Lebanon's climate-mitigation efforts through CCUS deployment.

This study represents the first integrated long-term CCUS modeling effort targeting the Lebanese offshore environment by simulating the representative geological conditions of the Levantine Basin and conducting a comprehensive cost assessment. It delivers a cost-benefit implementation roadmap, while pinpointing the optimal combinations of formation properties required for effective CO₂ sequestration. The findings not only provide a replicable framework but also offer valuable regional insights for developing countries exploring CCUS technologies as part of their carbon mitigation strategies.

Keywords: Carbon Capture Utilization and Storage, CCUS, Saline Aquifers, Offshore Lebanon, Sustainability, Levantine Region

Introduction

Climate change refers to long-term shifts in global weather patterns and temperatures. While these shifts were historically influenced by natural phenomena, such as major volcanic eruptions and variations in solar activity, the current trend is primarily driven by anthropogenic activities ([Venturini, 2022](#)). Climate change represents a significant global challenge, characterized by rising temperatures, increased frequency and intensity of extreme weather events, and profound impacts on ecosystems and human well-being ("Climate Change 2021—The Physical Science Basis," 2021).

The central challenge in addressing climate change is reducing atmospheric greenhouse gas (GHG) concentrations, with carbon dioxide (CO₂) being the most prevalent and significant contributor. Since the onset of the Industrial Revolution, particularly owing to fossil fuel combustion and large-scale industrialization, atmospheric CO₂ levels have increased by more than 50%, reaching levels significantly higher than pre-industrial concentrations ("Climate Change 2021—The [Physical Science Basis](#)," 2021). In 2018, fossil fuels accounted for approximately 89% of global CO₂ emissions ([Cao et al., 2020](#)). Among these, coal remains the most carbon-intensive, contributing an estimated 0.3°C to 1°C of the observed global temperature increase ("Climate Change 2021—The [Physical Science Basis](#)," 2021). Oil combustion releases a substantial volume of CO₂, accounting for approximately one-third of global cumulative CO₂ emissions ([ClientEarth, 2022](#)). In addition to its role in atmospheric carbon loading, the oil industry is associated with frequent spills that have severe and long-lasting ecological impacts on marine environments. As of 2024, the atmospheric concentration of CO₂ has reached approximately 420 parts per million (ppm), nearly double the pre-industrial level of around 280 ppm ([NASA, 2025](#)).

Carbon Capture, Utilization, and Underground Storage (CCUS) is increasingly being recognized as a critical mitigation strategy for reducing atmospheric CO₂ concentrations. The process involves capturing CO₂ emissions at their source, such as power plants and industrial facilities, followed by transportation and permanent storage in secure geological formations ([K. F. Bou-Hamdan et al., 2025](#)). CCUS is particularly relevant in hard-to-abate sectors, including heavy industry and energy production, where direct emission reductions are technologically or economically challenging. Among the storage options, saline aquifers are considered the most viable for large-scale deployment because of their widespread presence and high storage capacity.

Lebanon, located along the eastern Mediterranean coast, presents a promising opportunity for offshore CO₂ storage, particularly through its saline aquifer formations. The proximity of coastal industrial hubs and energy production facilities further enhances the potential for CCUS deployment. However, a thorough assessment is required to evaluate the feasibility and effectiveness of such a project within Lebanon's geological, economic, and environmental contexts. This study employs numerical simulations, along with a sensitivity study, to evaluate the best-case scenario for CO₂ storage, supported by economic and feasibility analysis. It models the injection of CO₂ into an offshore saline aquifer to assess the storage capacity, long-term integrity, and potential risks. Additionally, the technical aspects of implementing CCUS in Lebanon are examined, including cost estimates, potential funding sources, and incentives for industrial stakeholders. The study also evaluates the environmental benefits of reduced CO₂ emissions in Lebanon, with an emphasis on mitigating climate change, improving air quality, and fulfilling Lebanon's climate pledges.

This research evaluates the potential for CO₂ capture and storage in a saline aquifer off the coast of Lebanon. It presents a comprehensive analysis of CCUS implementation in the region, incorporating both the technical and economic dimensions. This study also examines the geological characteristics of Lebanon's offshore formations, providing a foundation for understanding the economic opportunities and challenges associated with CCUS in Lebanon.

Methodology

Availability Of Data

Understanding historical temperatures, sea levels, and biological evolution requires a thorough analysis of the Oligocene stratigraphy, which focuses on sedimentary rock layers from the Oligocene epoch (34-23 million years ago) (Miall, 2010). Due to limited local data, this research utilizes average data from the Levantine Basin, which encompasses the Lebanese offshore territory. The Oligocene strata in this region exhibit varied levels of porosity and permeability across multiple lithologies including sandstone, limestone, and shale (Glazer et al., 2024). The potential for gas reserves in this region is evidenced by nearby analogues such as the Tamar and Zohr gas fields, the latter of which demonstrates excellent reservoir quality despite the challenges posed by high salinity (Gao et al., 2019). For quantitative analysis, standard engineering calculations were applied and implemented to enable the estimation of CCUS costs under multiple scenarios specific to Lebanon.

Software Simulation

A numerical simulator was used in this study to model CO₂ injection and storage, including thermal effects. The model is governed by the following set of laws:

a. The molar balance for a phase is given by (Mehmani et al., 2012):

$$\frac{\partial}{\partial t} \left[\phi \sum_{(\beta)} (x^a_{\beta} S_{\beta}) \right] + \nabla \cdot \sum_{(\beta)} (x^a_{\beta} q_{\beta}) + \phi \sum_{(\beta)} (D_{\beta} \nabla x^a_{\beta}) = Q_i \quad (1)$$

where ϕ is the rock porosity, x^a_{β} is the mole fraction of component a (water or gas) in phase B (aqueous or vapour), S_{β} is the saturation of phase B, μ_{β} is the molar density of phase B, q_{β} is the Darcy velocity of phase B, D_{β} is the diffusion coefficient of phase B, and Q_i is a source term.

b. The Darcy velocity is given by (Whitaker, 1986):

$$q_{\beta} = (K k_{\beta} / \mu_{\beta}) \nabla (P_{\beta} - \rho_{\beta} g z) \quad (2)$$

where k is the saturated media permeability, k_{β} is the relative permeability of phase B, μ_{β} is the viscosity of phase B, P_{β} is the pressure of phase B, ρ_{β} is the mass density of phase B, g is the gravitational constant, and z is the height coordinate.

c. The energy balance is given by (Lovejoy et al., 2021):

$$\partial/\partial t \left[\phi \sum_{(\beta)} (s_{\beta} \eta_{\beta} U_{\beta}) + (1 - \phi) \rho_r c_r T \right] + \nabla \cdot \sum_{(\beta)} [q_{\beta} \eta_{\beta} H_{\beta} + \kappa \nabla T] = Q_e \quad (3)$$

where U_{β} is the internal energy of Phase B, ρ_r is the rock density, C_r is the rock heat capacity, k is the rock thermal conductivity, and Q_e is the source term.

After reviewing multiple sources and considering the requirements of the simulation software, the data and property values listed in Table 1 were compiled.

Table 1—Simulation Parameters

Parameter	Value
Section Dimensions	5000 m (horizontal), 400 m (vertical), top at 2000 m depth under 1500 m seawater
Injection Details	1 Mt/year of CO ₂ injected at 1000m depth for 6.85 years
Reservoir Saturation	Initially saturated with brine
Pressure Management	Producer at bottom layer operates at constant pressure to simulate deep aquifer and prevent pressure buildup
Rock Compressibility	4.35×10^{-5} 1/Bar
Porosity & Permeability	Dynamic
2500 days	When injection stops
1,825,000 days (5000 years)	End of simulation
Pressure	600 bars at 2000 m depth
Water Column Above	1500m
Initial CO ₂ in Brine	0

In this study, the simulation was conducted in two dimensions (2D), with the reservoir tops located at a depth of 3500 m. Porosity and permeability were treated as dynamic properties within the model.

The simulation scenario specifies two main time periods, one at 2500 days, corresponding to the end of the CO₂ injection, and the other at 1,825,000 days, when the simulation ends. The simulator is configured to output the fluid-in-place report and gas-phase partitioning data to quantify the solution trapping and residual trapping mechanisms.

Regarding the rock properties, a typical compressibility value of $4.35/10^5$ 1/bar is assigned, based on the estimated depth of formation where CO₂ injection is applied.

The hydrostatic pressure equilibration sets the initial reservoir pressure to 600 bar at a depth of 2000 m, below 1500 m of water. The initial CO₂ concentration in the brine was set to zero. A temperature-depth profile was defined using a temperature table to specify the initial thermal conditions.

Regarding well configuration, the simulation defines injector and producer wells based on IJK logical coordinates, specifying both the location and operational schedule. The gas injector was positioned at the center of the cross-section with a CO₂ injection target of 1 megatonne per year. After 2500 days, the injector was closed to flow. The producer well, operating with a bottom hole pressure of 120.5 bars, was completed in the bottom layer to avoid pressure buildup.

Economic and Feasibility Assessment

This economic study evaluates the feasibility of the CCUS project in offshore Lebanon. The approach utilizes recognized engineering standards and follows the structured framework presented in Figure 1.

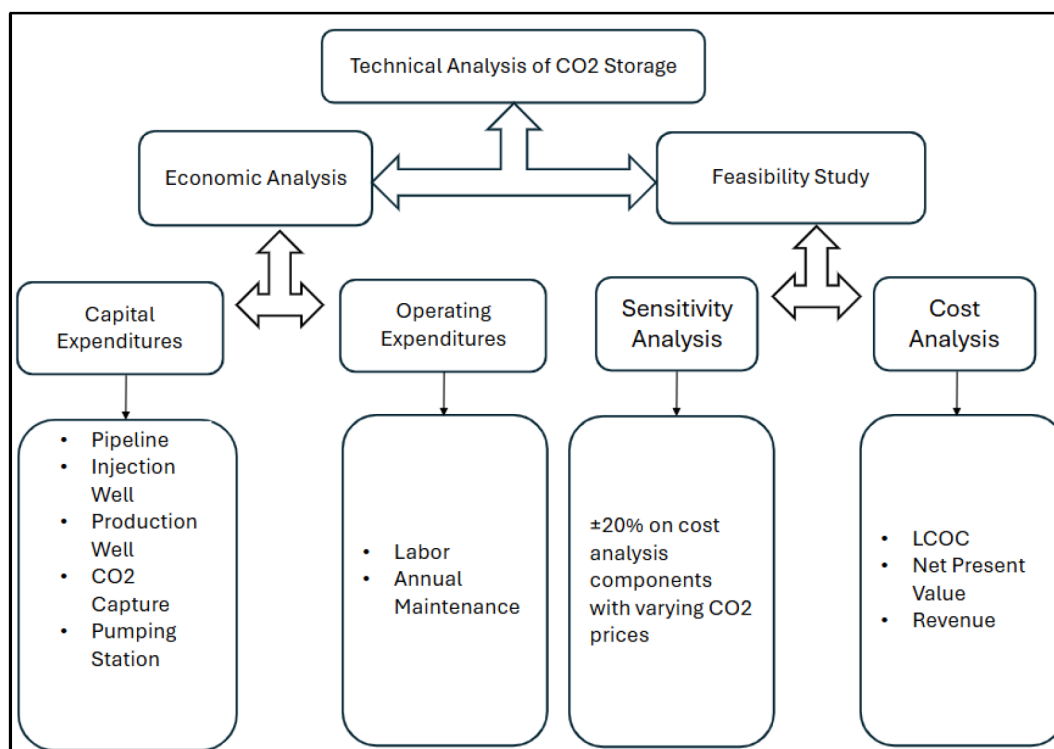


Figure 1—Flow Chart Representing Steps of CCUS Technical Assessment

The economic analysis is evaluated by determining Capital and Operating Expenditures. The Capital Expenditures (CAPEX) were estimated by considering several parameters, including the annual CO₂ emissions, required pipeline installations, pumping/compressor stations, and wells needed. Annual CO₂ emissions were gathered from data provided by the Lebanese Ministry of Energy and Water. The estimated costs were calculated using the following equations:

The carbon dioxide capture price was obtained using the following relation (Schmelz et al., 2020):

$$\text{Total CO}_2 \text{ Capture Price} = \Sigma \text{ Annual CO}_2 \text{ Emission} \times \text{Cost Per Ton} \quad (4)$$

The pipeline cost was obtained using the following relation:

$$\text{Pipeline Price} = (\text{Distance} \times \text{Carbon Steel Pipe Price}) + (\text{Coating Price} \times \text{Distance}) \quad (5)$$

Other costs, such as injector and production wells, were obtained based on the cost of an exploration well drilled in Lebanon (Block 4). The compressor station was estimated to match the Lebanese Market.

Operating Expenditures (OPEX) were also estimated by considering labor costs, monitoring costs, and maintenance costs. The costs considered the Lebanese market price along with international prices.

The feasibility assessment was evaluated by applying cost and sensitivity analyses. The cost analysis considered the levelized cost of carbon dioxide (LCOC), the net present value (NPV), and the expected revenues as follows:

The LCOC represents the average cost per ton of CO₂ captured over lifetime "t", accounting for both capital and operational expenses. The following relation can be used to calculate it (Scottish Government, 2013):

$$\text{LCOC} = \text{CAPEX} + \text{OPEX} / (\text{CO}_2 \text{ Emission in Project Lifetime}) \quad (6)$$

The other parameters computed are the net present value (NPV) and total revenues, which are commonly used to evaluate the profitability of a project. This study was concluded with a sensitivity analysis (± 20% variation) of some critical parameters such as CAPEX, OPEX, and CO₂ cost.

Results and Discussion

Simulation

The model was simulated under different conditions of porosity, permeability, and salinity to account for uncertainties in these parameters. The porosity values considered were 20, 25, and 30%. The permeability values were 100, 550, and 1000 mD. The salinity values were 35,000, 67,500, and 100,000 PPM. A total of 27 cases were simulated, and the results are discussed in the following text. [Table 2](#) presents some of the main observations for 3 different scenarios.

Table 2—Simulation's observations

Scenario	Porosity	Permeability (mD)	Salinity (PPM / Molality)	Main Observations
1	20%	100	35,000 / 0.621	Gas rises slowly. Residual gas around the injector well. High percentage of trapped gas.
2	25%	550	67,500 / 1.239	Faster gas migration. Little to no residual gas.
3	30%	1000	100,000 / 1.901	Very fast gas migration. No residual gas. Low amounts of trapped gas.

The results clearly show a direct relation between permeability and gas migration behavior. Low permeability values significantly reduce upward gas migration and favor gas trapping, as shown in the first scenario of [Table 2](#). This can be advantageous for long-term storage as it also lowers the chance of gas leakage. Conversely, the second and third scenarios, characterized by higher permeability values, exhibit more rapid gas migration, which may compromise storage security. Although porosity also affects the capacity of the reservoir to store gas, the storage efficiency is largely determined by the ability of the gas to be trapped and its migration rate.

The convection process, which is crucial for gas mixing and dissolution, appears to be significantly affected by formation fluid salinity. For instance, lower salinity promotes gas dissolution into the brine, facilitating long-term, secure storage. Higher salinity slows down the dissolution process and increases the possibility of mobile gas remaining in the system for longer periods of time.

For the considered cases, a combination of low salinity (35,000 PPM), moderate permeability (550 mD), and high porosity (30%) yielded the best storage conditions. This arrangement strikes a balance between gas movement and trapping efficiency to guarantee good storage capacity.

Regarding the effect of porosity on the rate of carbon dioxide dissolution, it has been shown that for a fixed value of permeability and salinity, the higher the porosity, the higher the rate of carbon dioxide dissolution. This is evident in [Figure 2](#), where the blue line represents a porosity of 30%, green line represents a porosity of 25%, and red line represents a porosity of 20%.

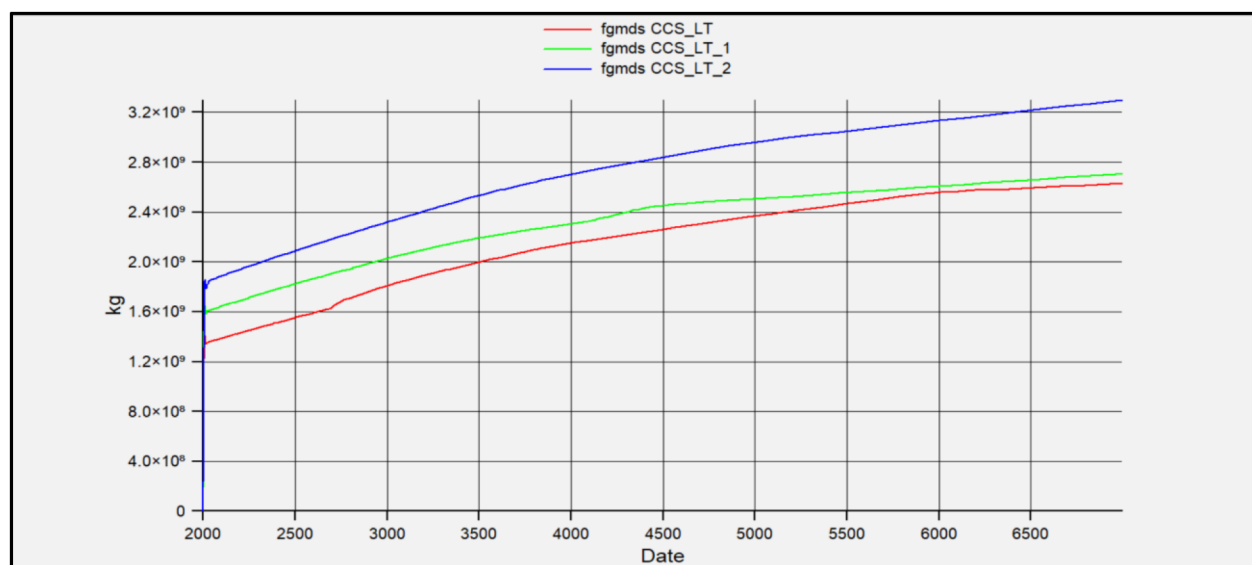


Figure 2—Dissolved gas at different porosity values.

Considering the best scenario, and after 5000 years, the amount of dissolved gas was almost 3×10^9 kg, covering only the red region in Figure 3. This outcome suggests that larger amounts of CO₂ can still fit into the aquifer.

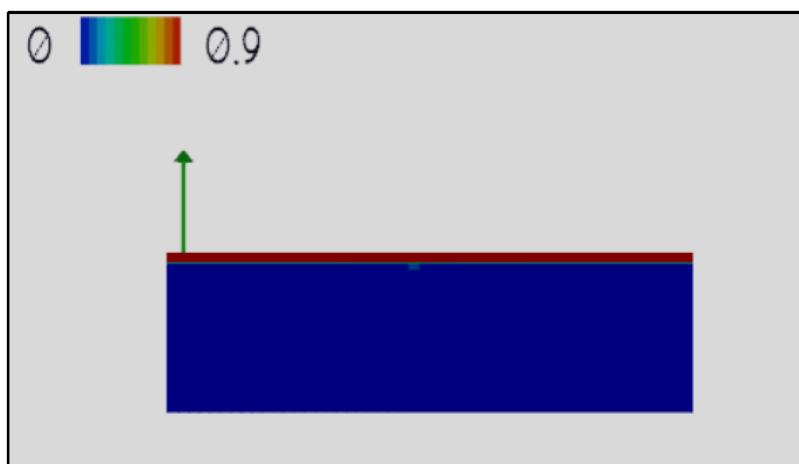


Figure 3—Dissolved amount of CO₂ in aquifer.

Technical Evaluation

Geological Assessment. Several criteria were considered to confirm which reservoir could be a potential candidate for carbon dioxide storage. Some of these are the permeability, porosity, presence of cap rock, and structural geometry. Both high porosity and high permeability are essential for successful CO₂ storage. The presence of a cap rock is crucial to ensure that CO₂ does not migrate outside the reservoir, and the structural geometry provides a good insight for detecting future leakages (Chen et al., 2025). Fields that are close to each other usually share similar geological characteristics. The Qana field, located in Block 9 offshore Lebanon, as shown in Figure 4, is approximately 10 km from the Karish Field (Anton Macfadyen, 2022). Moreover, Block 4 and Block 9 both belong to the Levantine Basin, so they share some similar geological layers (K. Bou-Hamdan, 2020).

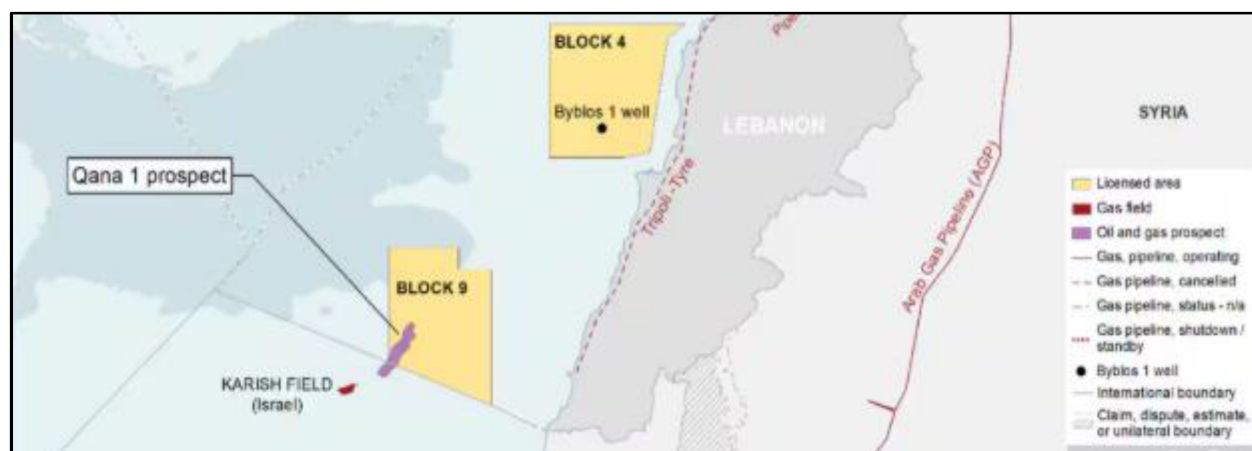


Figure 4—Exploration drilling activities in Offshore Lebanon (S&P Global, 2023)

The parameters that are considered when selecting the appropriate offshore location for CCUS, solely based on geological and geographical location, are the porosity, permeability, cap rock, salinity, and distance from the shore. In terms of porosity, permeability, and brine salinity, both the Qana and Byblos fields are expected to contain layers of sandstone and interbedded carbonates. Recent findings suggest that the sandstone layers in the Byblos field are thinner than those in Southern Lebanon. Regarding the cap rock, the exploration surveys and drilling activities in block 4 and 9 showed an approximately 1 km thick salt layer (K. Bou-Hamdan, 2020). These layers are impermeable, and therefore, are considered an effective barrier for carbon dioxide, thus ensuring containment. This assessment suggests that both blocks have similar geological advantages and may be suitable for CO₂ storage. However, Block 4 is located closer to the shore and manufacturing industries, so it is a more favorable location for CO₂ storage. Therefore, the Economic assessment in this study was based on this block.

Economic Assessment. In this section, the total cost of CCUS project implementation in Lebanon is discussed. Table 3 shows the amount of carbon dioxide emissions in tons per year for the five major power plants in Lebanon located near the shore.

Table 3—Existing Thermal Power Plants Data (Source Lebanese Ministry of Energy and Water)

Power Plant	CO ₂ Emissions (tonnes/year)
Jieh	854,430
Beddawi	1,986,925
Zahrani	1,986,925
Zouk	1,541,313
Tyre	221,727

The economic assessment considered the following parameters:

Distance:

Block number 4 is located around 30 km from the beach and is exactly opposing the center of Beirut (K. Bou-Hamdan, 2020). The distance from each power plant to Block 4 was estimated using geospatial tools, and the results are shown in Table 4.

Table 4—Distance from Block 4 to Each Power Plant

Power Plant	Land distance to Center of Beirut (km)	Total distance to Block 4 (km)
Jieh	28.69	41.51
Beddawi	78	83.57
Zahrani	46.73	55.53
Zouk	25.6	39.44
Tyre	73.2	79.11

Carbon Capture Cost:

For saline aquifers, specifically offshore, the cost of carbon capture per ton is around \$18 (Schmelz et al., 2020). Based on Table 3, the overall carbon capture cost for each power plant was computed, and the results are shown in Table 5. The estimated total carbon capture cost for the selected powerplants is around \$119,000,000.

Table 5—Total Carbon Capture Cost Calculation

Power Plant	CO ₂ Emission Per Year (tons/year)	Total Carbon Capture Cost (\$)
Jieh	854,430	15,379,740
Beddawi	1,986,925	35,764,650
Zahrani	1,986,925	35,764,650
Zouk	1,541,313	27,743,634
Tyre	221,727	3,991,086
Total		118,643,760

Pipeline Price:

The carbon steel pipe used in Carbon Capture and Storage (CCS) projects is of grade (ASME B36.10M) and has a weight of 79.7 kg/m (American Society of Mechanical Engineers, 2015). These pipes cost \$300/m and require fusion-bonded epoxy coating owing to carbon dioxide impurities (Trident Steel, 2025). Moreover, the epoxy coating costs \$30/ft, which is equivalent to \$98.43/m (Anup Kumar Dey, 2023). Table 6 presents the results obtained using Equation 5.

Table 6—Total Pipeline Infrastructure Cost

Power Plant	Distance (m)	Pipe Price (\$)	Epoxy Coating (\$/m)
Jieh	28,690	8,607,000	2,823,956.7
Beddawi	83,570	250,71,000	8,225,795.1
Zahrani	55,530	16,659,000	5,465,817.9
Zouk	39,440	11,832,000	3,882,079.2
Tyre	79,110	23,733,000	7,786,797.3
Total		85,902,000	28,184,446.2

Labor Cost:

Labor costs vary according to project complexity, location, and required expertise. It currently ranges from \$20 to \$150 per hour. For more accuracy, pipe welding assembly mean value is 0.70 man-hour which equates 3.63 cm per hour. Moreover, the pipe diameter for CSS projects is approximately 32 mm (Lu et al., 2019).

Due to the lack of data, \$30 was taken as the minimum labor wage in Lebanon for the following project. [Table 7](#) shows the forecasted labor costs under these conditions.

Table 7—Total Labor Cost Calculations

Power Plant	Man-hour per meter	Distance (m)	Man/hour	Labor cost (\$)
Jieh	19.28	28,690	553.1	16,594
Beddawi	19.28	83,570	1,611.2	48,336
Zahrani	19.28	55,530	1,070.6	32,118
Zouk	19.28	39,440	760.4	22,812
Tyre	19.28	79,110	1,525.2	45,757
Total			5,520.6	165,619

Injection Well:

As drilling operations commenced in Block 4, the cost of the drilling operation was reported to be approximately \$60,000,000.

Pipe Monitoring & Leakage:

For acoustic monitoring systems, pipe monitoring and leakage range from \$20,000 to \$100,000 per kilometer with an average of \$60,000. Estimating this value at \$45,000, the costs of monitoring and leakage maintenance are listed in [Table 8](#).

Table 8—Total Cost of Pipeline Monitoring and Leakage

Power Plant	Distance (km)	Pipeline Monitoring (\$)	Cost (\$)
Jieh	28.69	45,000	1,291,050
Beddawi	83.57	45,000	3,760,650
Zahrani	55.53	45,000	2,498,850
Zouk	39.44	45,000	1,774,800
Tyre	79.11	45,000	3,559,950
Total			12,885,300

Compressor Stations:

The cost for one facility might vary from thousands of dollars to millions of dollars, depending on the size of the and capacity of the station. Also, multiple stations might be needed to ensure the gas remains pressurized, aiding the transportation. The cost of all the compressor stations needed for this project can be roughly assumed to be around \$30,800,000.

Production Well:

Since it is a deep-water well, the production well is estimated to be between 60 and 120 million dollars ([Abed El Rahim & Al Husban, 2021](#)). In this study, it was assumed to be in the range of \$90,000,000. By combining all the costs, the estimated cost for the CCUS project in Lebanon is presented in [Table 9](#).

Table 9—Estimated Total Cost for CCUS Project

Component	Total Cost (\$)
Pipeline Price	85,902,000
Pipeline Monitoring	12,885,300
Carbon Capture Cost	118,643,760
Labor Cost	165,619
Injection Well	60,000,000
Annual Maintenance	21,360,000
Compressor Stations	30,800,000
Epoxy Coating	28,184,446.2
Production Well	90,000,000
Total	447,941,125.20

Feasibility Assessment. A feasibility study is essential for evaluating the economic, environmental, and technical factors that affect project feasibility, including key elements such as OPEX, CAPEX, carbon pricing, and storage. Thus, this detailed feasibility study is significant in evaluating the technical and financial challenges of CCUS in Lebanon. By evaluating factors such as LCOC and CO₂ pricing with sensitivity analysis, this feasibility study will be a key decision-making reference to ensure that this project is both environmentally and economically viable. These evaluations are critical in CCUS implementation for identifying optimal strategies and overcoming limitations.

Economic Assumption and Model

In this section, the financial model is developed based on economic assumptions. These are based on industry trends and available data for applied sensitivity analysis, financial metrics, and estimated costs.

CO₂ Pricing

The cost of carbon dioxide is a critical factor influencing the economic viability of this project. If CO₂ trade and carbon pricing are implemented, this will significantly improve the feasibility of the CCUS. Based on the European Union Emissions Trading Scheme (EU, ETS) and other carbon pricing frameworks, a proper carbon market can encourage investment in carbon reduction technologies (European Commission, 2019). Because of the lack of data, three scenarios were implemented by taking the minimum, average, and maximum prices of carbon dioxide stored per ton as 13, 40, and 75 dollars, respectively, to accurately evaluate all possible scenarios. By applying Equation 5, the total estimated CAPEX was computed, as shown in Table 10. OPEX was also computed, as shown in Table 11, and the annual maintenance costs were assumed to be around \$21,360,000 for the whole project.

Table 10—CAPEX Calculation.

CAPEX	
Component	Total Cost (\$)
Pipeline Price + Coating	114,086,446
Pipeline Monitoring	12,885,300
Carbon Capture Cost	92,304,845
Injection Well	60,000,000
Compressor Stations	30,800,000
Production Well	90,000,000

CAPEX	
Component	Total Cost (\$)
Total	400,076,591

Table 11—OPEX Calculation (Labor Costs assumed to become part of maintenance costs after first year).

OPEX		
Component	Total Cost (\$/year)	Total Cost (\$/5 years)
Labor Costs	165,619	165,619*
Annual Maintenance	21,360,000	106,800,000
Total	21,525,619	106,816,569

The LCOC was calculated using Equation 6 and a value of 8.22 was obtained.

Financial Model. The financial model of this project incorporates the possible revenue generated from selling the stored carbon dioxide and the associated OPEX and CAPEX. Although this case may or may not be applicable in this situation, it was considered in this study. The revenue generated was calculated based on the three projected prices of CO₂ as shown in Table 12.

Table 12—Revenue Calculated based on the projected CO₂ cost.

Cost of CO ₂	Revenue (\$)
Minimum Value \$13	64,161,541
Mid-Range Value \$40	242,127,181
Maximum Value \$75	472,823,381

NPV values were also computed, and the results are presented in Table 13.

Table 13—NPV and Annual Cash Flow Calculations

Cost of CO ₂ (\$)	Annual Net Cash Flow (\$)	$\frac{(1+r)^{-n}}{r}$	Capex (\$)	NPV (\$)
45,138,991.42	23,613,372.37	7.0236	400,076,591	-385,413,047
183,596,259.3	162,070,640.3	7.0236	400,076,591	587,055,420
363,077,902.9	341,552,283.9	7.0236	400,076,591	1,847,662,692

Sensitivity Analysis. The sensitivity analysis considers a ±20% variation in CAPEX and OPEX (1 year) to account for potential reductions or increases in the project costs. This is reflected in the costs, as follows:

- **Base Case CAPEX:** \$400,076,591.3, as calculated previously.
- **Reduction in CAPEX:** 20% lower, bringing the total CAPEX to \$320,061,273.04.
- **Increase in CAPEX:** 20% higher, bringing the total CAPEX to \$480,091,909.56.
- **Base Case OPEX:** \$21,525,619, as calculated previously for 1 year.
- **Reduction in OPEX:** 20% lower, bringing the total OPEX to \$17,220,495.20 per year.
- **Increase in OPEX:** 20% higher, bringing the total OPEX to \$25,830,742.8 per year.

The sensitivity also considers a $\pm 20\%$ variation in the amount of carbon dioxide, which is reflected by:

- **Base CO₂ Injection:** 6,591,320 tons per year.
- **Decrease:** 5,273,056 tons per year.
- **Increase:** 7,909,584 tons per year.

Results of Sensitivity Analysis

Based on the sensitivity analysis, carbon dioxide pricing is the most influential factor affecting a project's economic performance. At a low market rate with a minimum value scenario of \$13/ton, the NPV was - \$365,818,801.12, indicating significant financial risk. However, with a mid-price of \$40/ton and maximum of \$75/ton, the NPV was positive, thus yielding optimistic outcomes. This leads to the conclusion that strong carbon pricing is essential for CCUS feasibility. Based on CAPEX, a 20% increase negatively affects the economic outcome, whereas a 20% decrease positively improves the NPV, making the project more feasible. While OPEX price reduction by 20% improved the cash flow, its effect remained moderate in comparison to CO₂ pricing or CAPEX. However, this parameter becomes significant if project duration increases (in years). Lastly, based on the CO₂ injection volume, a 20% increase led to an increase in profitability. In conclusion, careful cost control, high carbon pricing, cost optimization, and efficient technical performance will lead to more feasible CCUS projects in the region.

Conclusion

The CCUS project in offshore Lebanon is estimated to be approximately ± 500 million dollars and is considered a small-scale project compared to other worldwide CCUS implementations. Nonetheless, this study shows that it is of substantial value when supported by proper economic and technical coherence. Economically, this initiative broadens Lebanon's energy transition through adequate job and skill development. From a feasibility standpoint, economic viability is achieved when carbon dioxide pricing exceeds capture and storage costs, demonstrating how this structure encourages future investments. Moreover, a reduction in CAPEX and OPEX with environmental funding is a key factor in maximizing profitability. Technically, for such an offshore saline formation, the storage potential is targeted at specific geological conditions such as high porosity, balanced salinity, and moderate permeability. However, there are still some limitations to consider, such as the lack of laws and policies that organize and support the implementation of such projects in Lebanon, insufficient geological data, and limited research. Moreover, the failure to invest in this project will result in a diminished energy transition. It is estimated that restricting the availability of carbon dioxide storage will necessitate a 40% increase in investments in the power and industrial sectors, which will have detrimental effects on the environment. To unlock the full potential of this CCUS project and encourage environmental initiatives, Lebanon must take strategic action to tackle these limitations.

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